# EXPERIMENT 8

*Configuring the Smart Campus Sustainability & Carbon Tracker with Role-Based Access Control (RBAC) in Zoho Creator*

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## Title

Design and Implementation of User Permissions, Roles, Role Hierarchy and Data Sharing for the Smart Campus Sustainability & Carbon Tracker in Zoho Creator.

## Aim

To build a complaint-tracking module in the Smart Campus Sustainability & Carbon Tracker application in Zoho Creator using a structured 13-field complaint form (Sustainability_Complaint_Form) and report (Sustainability_Complaint_Report), and to implement granular role-based access control through a three-level role hierarchy and a custom Read Only data-sharing rule.

## Objectives

- To design Sustainability_Complaint_Form with 13 Basic Fields grouped into four layout panels: Identity, Contact, Complaint Details and Scheduling.
- To create Sustainability_Complaint_Report and use the Sustainability Dashboard showing complaints, resolution ratio and status-wise and category-wise workload.
- To define the role hierarchy Sustainability Officer → Facility Coordinator → Data Entry Staff.
- To configure a data-sharing rule from Facility Coordinator to Data Entry Staff with Read Only access.
- To understand how RBAC separates data ownership from data visibility.

## Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Smart Campus Sustainability & Carbon Tracker application created in Zoho Creator
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

## Introduction / Theory

### User Management in Zoho Creator

User management decides who can access an application and what each person may see and do within it. Users are added to an application and associated with permission sets (profiles) and roles; together these control access to forms, reports, pages and records. In the Settings area, the User Permissions section groups these controls under three tabs: Permissions, Roles and Data Sharing.

### Role-Based Access Control (RBAC)

RBAC is an access-control model in which rights are assigned to roles instead of individual users, and users acquire rights through the roles they hold. It simplifies administration, supports the principle of least privilege, improves data security and keeps the access policy easy to audit. On a campus this matters

because complaint records contain names and e-mail addresses (personal data) while energy, water and waste records drive the sustainability reports, so each group of users should see only what it needs.

## Key Concepts

| Concept | Description |
|---|---|
| Users | Individuals added to the application who sign in to work with its forms, reports and data. |
| Permissions | Named permission sets (profiles) defining what a user may do with components. This application uses Administrator, Write, Read and Customer. |
| Roles | Positions in the organisational structure (Sustainability Officer, Facility Coordinator, Data Entry Staff) that decide a user's place in the hierarchy and the visibility of records. |
| Role Hierarchy | A tree in which each role reports to another. A higher role can generally view the records of roles beneath it; the reverse does not apply. |
| Data Sharing | Share rules that extend record visibility beyond the hierarchy, from one role or user to another, with a defined level of access (Read Only or Read/Write). |

## Application Specifications

| Parameter | Detail |
|---|---|
| Application Name | Smart Campus Sustainability & Carbon Tracker |
| Form Name | Sustainability_Complaint_Form |
| Report Name | Sustainability_Complaint_Report |
| Form Architecture | 13 Basic Fields in 4 layout panels (Identity, Contact, Complaint Details, Scheduling) |
| Role Hierarchy | Sustainability Officer → Facility Coordinator → Data Entry Staff |
| Custom Sharing Rule | Facility Coordinator → Data Entry Staff (Read Only) |

## Form Field Design

| Panel | Field Name | Field Type |
|---|---|---|
| Identity | Reporter_Name | Name |
| Identity | Reporter_Type | Radio (Student, Staff, Visitor) |
| Identity | Reporter_ID | Single Line |
| Contact | Contact_Number | Phone |
| Contact | Email_Address | Email |
| Contact | Campus_Address | Address |
| Complaint Details | Complaint_Category | Drop Down (Water Leakage, Excess Electricity, Waste Overflow, Plastic Waste, Broken Lights, E-Waste, Other) |
| Complaint Details | Date_Of_Observation | Date |
| Complaint Details | Estimated_Wastage | Number |
| Complaint Details | Resources_Affected | Multi Select |
| Complaint Details | Complaint_Description | Multi Line |
| Scheduling | Preferred_Inspection_Time | Time |
| Scheduling | Declaration_Of_Authenticity | Checkbox |

## Procedure

**1.** Sign in to Zoho Creator. The home page lists the available applications. Locate the Smart Campus Sustainability & Carbon Tracker application (Figure 8.1).

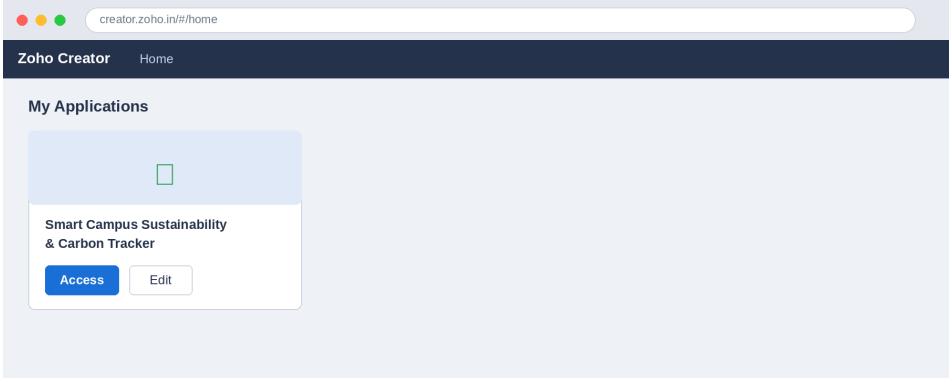



*Figure 8.1: Zoho Creator Home – Smart Campus Sustainability & Carbon Tracker (illustration)*

**2.** Open the application to view its dashboard, which contains the Sustainability_Complaint_Report, the KPI tiles and the energy and complaint charts (Figure 8.2).

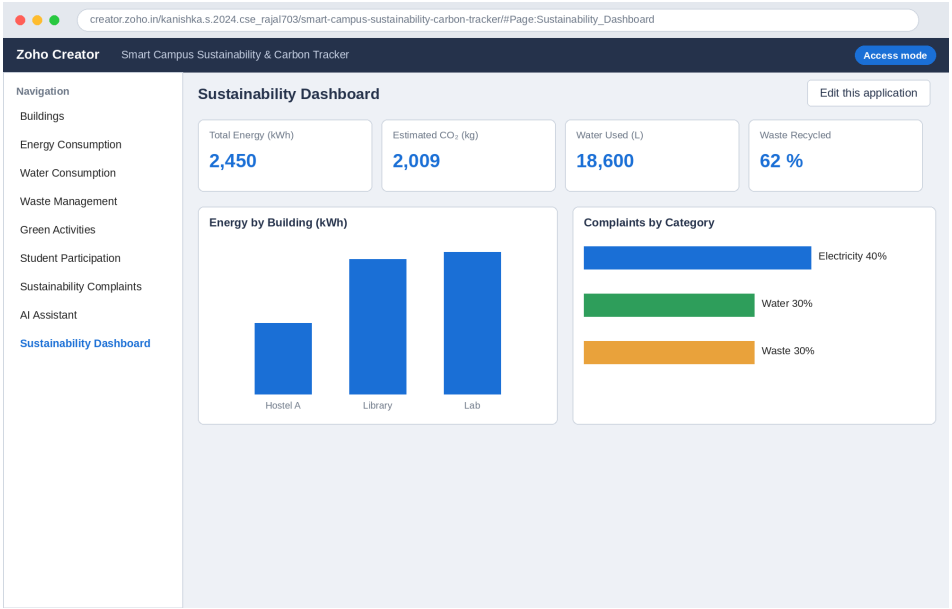



*Figure 8.2: Smart Campus Sustainability & Carbon Tracker – Access Mode (illustration)*

**3.** Enter Edit mode by selecting Edit on the application card (Figure 8.1) or Edit this application at the top-right (Figure 8.2).

**4.** Select the Settings tab from the top navigation bar (Design, Workflow, Settings).

**5.** Under the PERMISSIONS group, select User Permissions. The page is displayed with the tabs Permissions, Roles and Data Sharing.

**6.** Study the Permissions tab. The permission sets Administrator, Write, Read and Customer are listed. An Add Permission button is also provided (Figure 8.3).

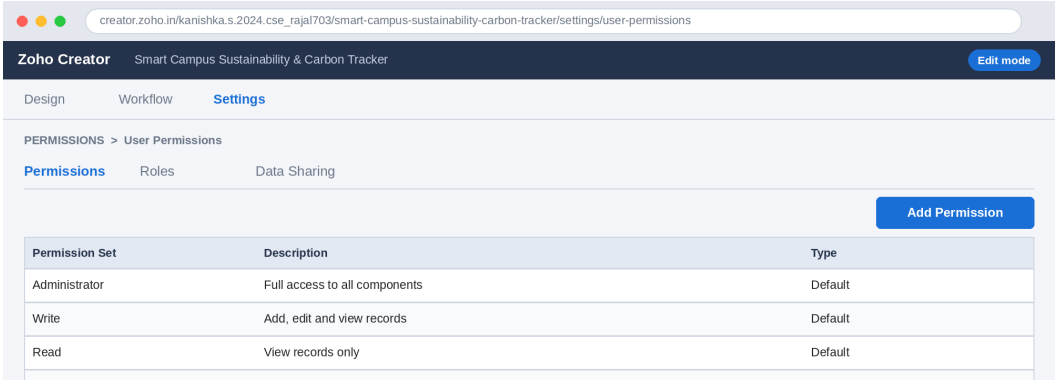



*Figure 8.3: User Permissions – Permissions Section (illustration)*

**7.** Select the Roles tab and arrange the role tree as Sustainability Officer → Facility Coordinator → Data Entry Staff. The Expand All, Collapse All, Manage Role Hierarchy and Add Role options are available (Figure 8.4).

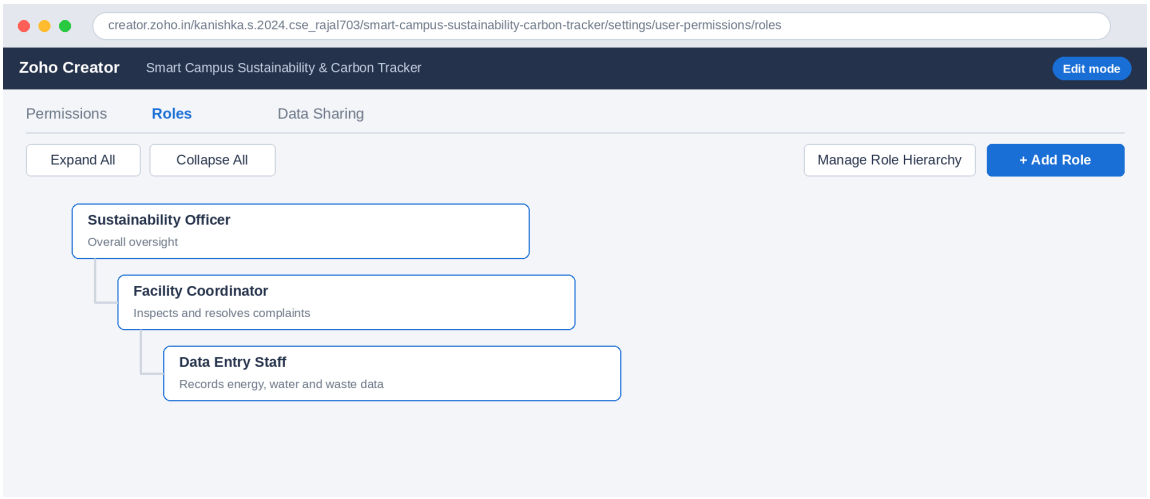



*Figure 8.4: Roles and Role Hierarchy (illustration)*

**8.** Select the Data Sharing tab and choose New Share Rule. Enter Name: Sustainability_Complaint_Form; Form: Sustainability Complaint; Records shared from: Roles. The dropdown offers Roles, Roles and Subordinates, and Users (Figure 8.5).

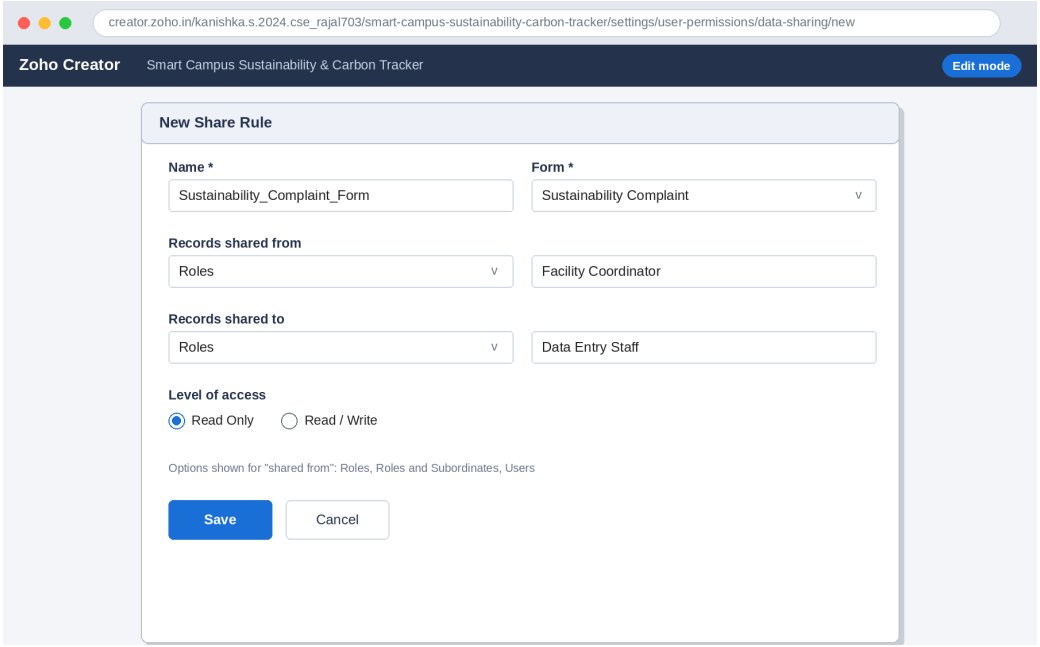



*Figure 8.5: New Share Rule – Rule Configuration (illustration)*

**9.** Set Records shared from → Roles → Facility Coordinator; Records shared to → Roles → Data Entry Staff; Level of access → Read Only.

**10.** Select Save. The rule appears in the Data Sharing list and can be reopened with Edit Sharing Rule (Figure 8.6).

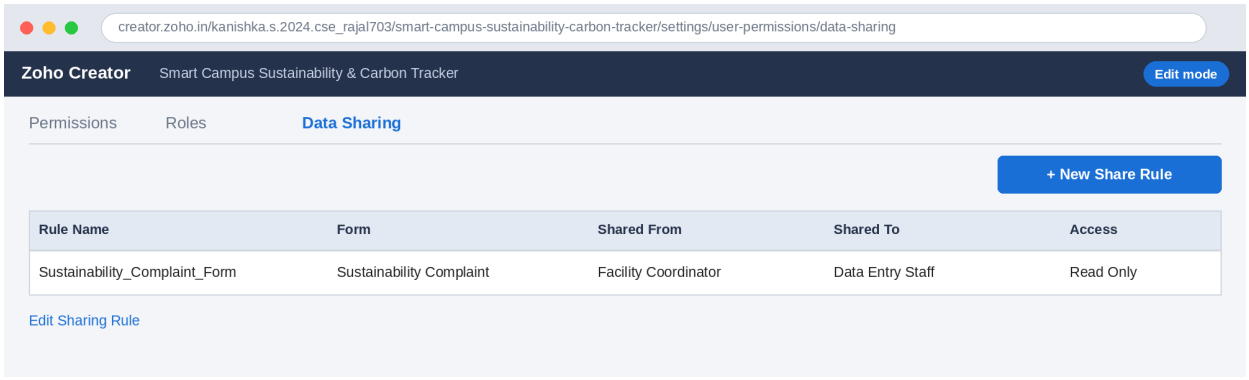



*Figure 8.6: Configured Data Sharing Rule (Facility Coordinator → Data Entry Staff, Read Only) (illustration)*

## Data Sharing

The application follows a layered model: Form (Sustainability_Complaint_Form) → Report (Sustainability_Complaint_Report) → Sustainability Dashboard → Security Layer (Permissions, Roles, Data Sharing). The role hierarchy runs from the Sustainability Officer (overall oversight) to the Facility Coordinator (inspection and resolution of complaints) and then the Data Entry Staff (day-to-day capture of energy, water and waste data). By default the hierarchy gives only upward visibility: higher roles see the records of the roles below them, but not the reverse.

The custom data-sharing rule shares Sustainability_Complaint_Form records from the Facility Coordinator role to the Data Entry Staff role at Read Only level. Read Only preserves data security because Data Entry Staff can see the complaint details they need for follow-up (location, category, priority and status) but cannot edit or delete them, so the records handled by the Facility Coordinator remain the single source of truth. It enforces least privilege by granting visibility without modification rights, keeps the scope

directional (Coordinator to Staff) so access never flows upward or sideways, and leaves a single named rule in the Data Sharing list that documents who can see what, and at what level, making the policy easy to audit.

### RBAC Working

| Level | Role | Intended Function |
|---|---|---|
| 1 (Top) | Sustainability Officer | Overall authority; able to view the records of roles reporting to it and to review the Sustainability Dashboard. |
| 2 | Facility Coordinator | Inspects and resolves complaints; owns the complaint records. |
| 3 | Data Entry Staff | Records energy, water and waste data; views shared complaints read-only. |

- Users in the Data Entry Staff role work with records according to their permission set.
- Because Data Entry Staff reports to the Facility Coordinator, who reports to the Sustainability Officer, higher roles can view the records of the roles below them.
- Lower roles do not automatically gain access to higher-role records; the Read Only share rule grants it in a controlled way.
- The permission sets (Administrator, Write, Read, Customer) further limit the operations each user may carry out.

## Result

The following were carried out in the Smart Campus Sustainability & Carbon Tracker application:

- The Smart Campus Sustainability & Carbon Tracker application, with Sustainability_Complaint_Form and Sustainability_Complaint_Report, was opened and its dashboard viewed.
- The permission sets Administrator, Write, Read and Customer were studied in User Permissions.
- The role hierarchy Sustainability Officer → Facility Coordinator → Data Entry Staff was created.
- A data-sharing rule from Facility Coordinator to Data Entry Staff with Read Only access was configured and saved.

## Conclusion

The experiment demonstrated how permissions, roles, role hierarchy and data sharing together form a role-based access-control mechanism in Zoho Creator. The three-level hierarchy and the directional Read Only share rule keep information channels isolated: Data Entry Staff see only what they need and cannot alter complaint records, while the Sustainability Officer and Facility Coordinator retain oversight. The result is a secure, auditable and scalable access model for the campus sustainability application.